

# Diagonal Pair Coulomb Density Matrix on a Sphere: a numerical study

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We tabulate, on the diagonal, the  $n$ th power of the primitive approximation for the pair Coulomb density matrix on a sphere, with up to  $n = 5$ , and we discuss its behavior at particles contact as  $n$  grows. Our results show that even on a curved surface quantum statistical physics is fundamentally different from its classical counterpart. If in the latter one needs to artificially introduce a regularization of the divergent attractive Coulomb potential preventing particles contact, for example with a hard core, in the former this is not necessary.

Keywords: Pair Coulomb Density Matrix; Primitive Approximation; End-Point Approximation; Sphere; Cusp Condition

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## I. INTRODUCTION

In a quantum many body (Monte Carlo) simulation of a Coulomb liquid it is often useful to have access to the low temperature pair Coulomb density matrix  $\rho$ . In the high temperature classical limit the *primitive approximation*  $\rho_{\text{primitive}}$  becomes exact. Within Path Integral Monte Carlo (PIMC) of a many body system [1] in order to access the quantum regime with less imaginary time discretization error it is important to use the *pair-product inter-action*. The imaginary time  $t$  is related to the temperature  $T$  of the liquid by  $t = \beta = 1/k_B T$  with  $k_B$  Boltzmann constant. The main ingredient to construct the pair-product inter-action is the inter-action of the pair density matrix or the *pair-inter-action*. Starting from  $\rho_{\text{primitive}}$  at a high temperature  $T_0$  we can halve the temperature simply by squaring it <sup>1</sup> [2, 3] (see also section IV.G of Ref. [1]). This lower temperature pair density matrix already possesses different

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<sup>1</sup> The squaring method can be used if an exact solution is known, so that successive squarings allow to reach temperatures  $T_0/2^n$  for  $n = 1, 2, 3, \dots$

properties than the primitive approximation. In position representation, on the diagonal

$$\left| \frac{\phi(\mathbf{r}_1, \mathbf{r}_2)}{k_B T_0} \right| = \left| \ln \left[ \frac{\rho_{\text{primitive}}(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)}{\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)} \right] \right| \xrightarrow{\mathbf{r}_1 \rightarrow \mathbf{r}_2} \infty, \quad (1.1)$$

where  $\phi(\mathbf{r}_1, \mathbf{r}_2)$  is the pair Coulomb potential between particle 1 at  $\mathbf{r}_1$  and particle 2 at  $\mathbf{r}_2$ . On the other hand, for a repulsive Coulomb potential, when Trotter formula holds [4], we can reach the temperature  $T_0 = T_n/n$  sending  $n \rightarrow \infty$  and  $T_n \rightarrow \infty$ , i.e. through the *continuum limit*<sup>2</sup>. In this case we find

$$|P(\mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)| = \lim_{n \rightarrow \infty} \left| \ln \left[ \frac{[\rho_{\text{primitive}}]^n(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; n/k_B T_n)}{[\rho_0]^n(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; n/k_B T_n)} \right] \right| \xrightarrow{\mathbf{r}_1 \rightarrow \mathbf{r}_2} \text{finite}, \quad (1.2)$$

where we denote with  $[\rho]^n$  the  $n$ th power of the pair density matrix:  $n = 2$  for the *square*,  $n = 3$  for the *cube*, and so on. Moreover a *cusp condition* must hold for  $P$  as the two particles approach each other, like in the flat space case [1, 5, 6].

The same contact value smearing property holds at any temperature  $T_0$ . This property makes it of fundamental importance not to use the primitive approximation for example in a simulation of an electron proton binary mixture where the divergence of the attractive Coulomb potential between unlike species at contact would inevitably let the fast electrons stick onto the slow protons, and consequently spoil the simulation. In other words quantum statistical physics does not need hard core regularization which is otherwise an essential but artificial feature in classical statistical physics necessary to avoid unlike particles collapse.

In this work we will show how this purely quantum smearing effect can be accomplished on a Sphere by approaching the  $n \rightarrow \infty$  continuum limit. We will not give the exact solution for the pair Coulomb density matrix as was done in flat space in Refs. [6–9] but we will give an approximate numerical solution instead starting from the classical high temperature limit primitive approximation. If we were able to find an exact solution of the pair Coulomb density matrix on a sphere we would not need to worry about deriving it from the primitive approximation. In absence of an analytic solution here we propose to use the path integral discretized in imaginary time with timestep  $\tau = 1/k_B T_n$  to reach a temperature  $T_0 = 1/nk_B \tau$  approaching the continuum limit  $\tau \rightarrow 0$ . Of course this limit cannot be reached numerically but we will be able to say something about the trend as  $n$  increases.

We will only worry about the *end-point approximation* (see section IV.F of Ref. [1]) to the pair Coulomb density matrix, i.e. the diagonal component of the pair-inter-action. Even if this is exact in the Euclidean three dimensional space due to the peculiar properties of the Coulomb pair potential [6] it becomes just an approximation on the Sphere.

## II. APPROACH TO THE CONTINUUM FOR THE PAIR COULOMB DENSITY MATRIX

We show a method to lower the temperature of the hot Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work in flat (Euclidean) open (with open boundary conditions) space of Refs. [6, 8, 9] to a curved surface.

We will denote with a Greek index a coordinate component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius  $a$  is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}. \quad (2.1)$$

with  $g(\mathbf{r}) = \det||g_{\alpha\beta}|| = a^4 \sin^2 \theta$ . We use polar coordinates  $\mathbf{r} = (r^1, r^2) = (\theta, \varphi)$  with  $\theta \in [0, \pi]$  the polar angle and  $\varphi \in [0, 2\pi[$ , so that  $\theta = 0$  is the north pole.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem (in Appendix B we treat the case when one of the two particles has infinite mass). Therefore, in full generality, we will study the position representation  $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  of the density matrix  $\rho = \exp(-\mathcal{H}t)$ , where  $\mathcal{H}$  is the Hamiltonian operator for the two bodies and  $t$  is the imaginary time that in statistical physics is identified with the inverse temperature  $\beta = 1/k_B T$  with  $k_B$  Boltzmann constant. These are initially at  $\mathbf{r}_1$  and  $\mathbf{r}_2$  and at an imaginary time  $t$  later they are at  $\mathbf{r}'_1$  and  $\mathbf{r}'_2$ .

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<sup>2</sup> For any finite  $n$  one can treat the attractive Coulomb interaction as well by removing the divergence at contact with a proper cutoff.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = -\mathcal{H} \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (2.2)$$

$$= \left( \frac{\lambda_1}{\sqrt{g(\mathbf{r}_1)}} \nabla_{1\alpha} \sqrt{g(\mathbf{r}_1)} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g(\mathbf{r}_2)}} \nabla_{2\alpha} \sqrt{g(\mathbf{r}_2)} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (2.3)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g(\mathbf{r}_1)g(\mathbf{r}_2)}, \quad (2.4)$$

where  $\mathcal{H}$  is the Hamiltonian of the system of the two particles  $i = 1, 2$ ,  $\lambda_i = \hbar^2/2m_i$  with  $m_i$  the mass of particle  $i$ ,  $g(\mathbf{r}_i) = a^4 \sin^2 \theta_i$ ,  $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial \theta_i, \partial/\partial \varphi_i)$  is the gradient respect to the coordinates  $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$ ,  $\delta$  is a two dimensional Dirac delta function,  $\rho$  is the pair density matrix, and  $\phi$  is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (2.5)$$

where  $Z_i e$  is the charge of particle  $i$  and for the distance between the two points  $\mathbf{r}_1$  and  $\mathbf{r}_2$ ,  $d(\mathbf{r}_1, \mathbf{r}_2)$ , on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = a \arccos[\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (2.6)$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2a \sin(d_g/2a). \quad (2.7)$$

For particles living **in** the surface [10] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -\frac{Z_1 Z_2 e^2}{2} \ln[\tan(d_g/2a)]. \quad (2.8)$$

In fact the Green function of Poisson equation in the Sphere surface,  $\nabla_{1\alpha} \sqrt{g(\mathbf{r}_1)} \nabla_1^\alpha G(\mathbf{r}_1, \mathbf{r}_2) = -2\pi \delta(\theta_1 - \theta_2) \delta(\varphi_1 - \varphi_2)$ , is  $G = -\ln[\tan(d_g/2a)]/2$  as is shown in Appendix A <sup>3</sup>. We will call  $\phi_{\text{on}}^g$ ,  $\phi_{\text{on}}^e$ ,  $\phi_{\text{in}}$  the three cases just illustrated. We mention that a parallel distinction can be made between the extrinsic and intrinsic curvature of a Sphere. We just recall that the relationship between the intrinsic (scalar) curvature  $R$  and the extrinsic curvature  $H$  of the Sphere of radius  $a$  is  $R = 2H/a = 2/a^2$ .

We will measure energy in units of the Hartree  $e^2/a_B$ , with  $e$  the charge of the electron and  $a_B = \hbar^2/m_e e^2$  Bohr radius, with  $m_e$  the mass of the electron. And length in units of the Bohr radius  $a_B$ . We will call  $\mu_i = m_i/m_e$  the reduced mass of particle  $i$ . Temperatures are in units of  $\hbar^2/m_e a_B^2 k_B$ .

The free particles,  $\phi \rightarrow 0$ , pair density matrix can be determined from the work of Bastianelli and Corradini [13]. We will call their solution for the single free particle  $i$ ,  $\rho_{i0}(\mathbf{r}_i | \mathbf{r}'_i; t)$ . Therefore the pair free density matrix will be

$$\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_{10}(\mathbf{r}_1 | \mathbf{r}'_1; t) \rho_{20}(\mathbf{r}_2 | \mathbf{r}'_2; t). \quad (2.9)$$

Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (2.10)$$

where  $P$  is the *pair-inter-action* [1].

Aim of this work is to tabulate  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  for the three Coulomb cases illustrated above,  $P_{\text{on}}^g$ ,  $P_{\text{on}}^e$ ,  $P_{\text{in}}$ . In the hot case, for  $\ln t \ll 1$ , for each of these cases, the *primitive approximation* [1]

$$P_{\text{primitive}} \approx t\phi, \quad (2.11)$$

holds.

Therefore, introducing a timestep  $\tau$ , we can find the PCDM at all subsequent timesteps through the following convolution path integral

$$\begin{aligned} \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; n\tau) &= \int \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_{1,1}, \mathbf{r}_{2,1}; \tau) \rho(\mathbf{r}_{1,1}, \mathbf{r}_{2,1} | \mathbf{r}_{1,2}, \mathbf{r}_{2,2}; \tau) \cdots \rho(\mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1} | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \times \\ &\quad \prod_{k=1}^{n-1} \sqrt{g(\mathbf{r}_{1,k})g(\mathbf{r}_{2,k})} d\mathbf{r}_{1,k} d\mathbf{r}_{2,k}, \end{aligned} \quad (2.12)$$

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<sup>3</sup> One may also choose to neutralize each charge with a uniform surface charge density of opposite charge in which case he should use  $-\ln[\sin(d_g/2a)]$  instead [11, 12]. Note that while charge neutrality is a necessary condition for thermodynamic stability in a non compact space it is just another possibility in a compact space like the sphere.

where we denote with  $\mathbf{r}_{i,k}$  the position of particle  $i$  at timeslice  $k$ ,  $d\mathbf{r} = dr^1 dr^2 = d\theta d\varphi$ , and we choose  $\tau$  small enough so that

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \approx \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2} \quad (2.13)$$

$$= \rho_{10}(\mathbf{r}_1 | \mathbf{r}'_1; \tau) \rho_{20}(\mathbf{r}_2 | \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2}, \quad (2.14)$$

is a good approximation. We will perform the  $4(n-1)$  integrations in Eq. (2.12) through a simple Monte Carlo (MC) quadrature [14] where a MC point is the tuple  $(\mathbf{r}_{1,1}, \mathbf{r}_{2,1}, \dots, \mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1})$ . It is important to stress that this is only possible if  $Z_1 Z_2 > 0$  otherwise the Coulomb potential will be unbounded from below and the path integral of Eq. (2.12) breaks down [4].

According to the analysis of Ref. [13] we can further approximate

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i | \mathbf{r}'_i; \tau) g^{1/4}(\mathbf{r}'_i) \approx \frac{\mu_i}{2\pi\tau} \exp \left\{ -\frac{\mu_i}{2\tau} \delta_{\alpha\beta} (\mathbf{x}_i - \mathbf{x}'_i)^\alpha (\mathbf{x}_i - \mathbf{x}'_i)^\beta - \tau \frac{V_R(x_i) + V_R(x'_i)}{2} \right\}, \quad (2.15)$$

where  $\delta_{\alpha\beta}$  is the Kronecker delta, the normalization  $2\pi\tau/\mu_i$  is just the one of a free particle of mass  $\mu_i$  on a plane and corresponds to the exact path integral performed with an action  $\int_0^\tau dt' \mu_i \delta_{\alpha\beta} \dot{x}_i^\alpha \dot{x}_i^\beta / 2$ . Here  $\mathbf{x} = (x^1, x^2)$  are the Riemann normal coordinates which are related to the polar coordinates  $\mathbf{r} = (r^1, r^2) = (\theta, \varphi)$  through

$$x^1 = ar^1 \cos r^2, \quad (2.16)$$

$$x^2 = ar^1 \sin r^2, \quad (2.17)$$

$x = \sqrt{\delta_{\alpha\beta} x^\alpha x^\beta}$ , and  $V_R$  is the effective potential due to the scalar curvature of the Sphere,  $R = 2M^2$  with  $M = 1/a$ , namely

$$V_R(x) = -\frac{M^2}{6} - \frac{5(Mx)^2 - 3 + [(Mx)^2 + 3] \cos(2Mx)}{48x^2 \sin^2(Mx)}, \quad (2.18)$$

which is equation (2.32) of Ref. [13].<sup>4</sup> This effective potential diverges to minus infinity for  $Mx = \theta = \pi$ . This divergence will produce an exponential divergence in Eq. (2.15) which can be numerically prevented through a cutoff. This procedure should not alter significantly the main picture since it amounts to flattening a neighbor of the south pole of the sphere.

We may additionally introduce the following exact Feynman-Kac expression for the one free body density matrix at time  $t$

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i | \mathbf{r}'_i; t) g^{1/4}(\mathbf{r}'_i) = \langle e^{-S_R} \rangle, \quad i = 1, 2 \quad (2.19)$$

$$S_R = \int_0^t dt' V_R(x_i) \quad (2.20)$$

where  $\langle \dots \rangle$  denotes an average over all single particle Gaussian random walks from  $\mathbf{r}_i$  to  $\mathbf{r}'_i$  in a time  $t$ . In Appendix C we give the diagonal part  $\rho_{i0}(\mathbf{r}_i | \mathbf{r}_i; t) = \rho_{i0}(\mathbf{0} | \mathbf{0}; t)$  up to orders  $t^8$ . Our Eq. (C1) coincides with the perturbative expansion (3.23) of Ref. [13] for  $d = 2$  spatial dimensions.

Even if in general we will be interested in the full  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  in practice in a many body simulation we may use the *end point approximation* for the inter-action as shown in section IV.F of Ref. [1] where one only needs the diagonal components  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = P(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t)$ . On the sphere the geodesic distance  $d_g(\mathbf{r}_1, \mathbf{r}_2)$  between any two points  $\mathbf{r}_1$  and  $\mathbf{r}_2$  is bounded from above by  $a\pi$ . Due to the Sphere symmetry, without loss of generality, we may choose  $\mathbf{r}_1 = (0, 0) = \mathbf{0}$  and  $\mathbf{r}_2 = (\theta, 0)$ , so that  $d_g = a\theta$ .

On the diagonal we will have

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t) = \rho_0(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t) e^{-P(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t)}, \quad (2.21)$$

we will then tabulate

$$P(m\delta; n\tau) = -\ln \left[ \frac{\rho(m\delta; n\tau)}{\rho_0(m\delta; n\tau)} \right], \quad n = 1, 2, 3, \dots, \quad m = 1, 2, 3, \dots, m_{\max} \quad (2.22)$$

for given small  $\tau$  and  $\delta$  so that  $m_{\max} = \pi/\delta$ . For an imaginary time  $t_n = n\tau$  this will require a number of  $4(n-1)$  quadratures for each  $m$  to calculate  $\rho(m\delta; n\tau) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t_n) = \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$  and  $\rho_0(m\delta; n\tau) =$

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<sup>4</sup> This idea to map the fluid on the curved manifold with an equivalent one on Euclidean space but subject to an effective external potential due to the effect of the curvature has been used before even in the classical limit [15].

$\rho_0(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t_n) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$  for  $m = 1, 2, 3, \dots, m_{\max}$ , through the corresponding path integral convolution (2.12). Each convolution requires 4 quadratures over the azimuthal angle  $\varphi \in [0, 2\pi[$  and the polar angle  $\theta \in [0, \pi]$  of each particle. In any case, in a many body simulation the pair-product inter-action is needed at a fixed small  $t$ , which determines the discretization error [1]. And one needs to tabulate the PCDM just once, at the beginning of the simulations.

In Fig. 1 we plot the ratio  $R(m\delta; t) = P(m\delta; t)/t\phi(m\delta)$  as a function of  $m$  for  $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2, \delta = 0.03$  and  $n = 2, 3, 4, 5, 6, 7$  in the top panel,  $n = 2$  in the other two panels. In the top panel, for the cases  $n = 6, 7$  we show just a few points from which we may hint the whole behavior. We see how at large particles distance the ratio  $R$  approaches 1 for  $n \leq 4$  but at contact it tends to a value below 1, approximately  $\approx 1/n$  as shown in Table I. In between,  $R_{\text{on}}^g \approx R_{\text{on}}^e$ , have a first peak at  $(\bar{d}_g(n), \bar{R}_{\text{on}}^g(n))$ . Where  $\bar{R}_{\text{on}}^g(n)$  increases with  $n$  and  $\bar{d}_g(n)$  remains constant for  $n \leq 4$  and grows for  $n > 4$  with an exception for  $n = 6$ . This is in agreement with the fact that each convolution in the path integral (2.12) involves 4 variables (two angles for the first particle and two for the second) and 4 convolutions will exhaust all 4 integrations. So that for  $n \leq 4$   $R$  will be smooth throughout the whole sphere. For  $n > 4$ ,  $R(d_g/a; t)$  have kinks within  $0 < d_g/a < \pi$  due to the divergence of the Coulomb potential at contact. The behavior of the first peak as a function of  $n$ , suggests that in the continuum limit  $n \rightarrow \infty$  the first peak will reach  $d_g/a = \pi$ , all the kinks will be washed out, and the ratio  $R$  will be smooth and monotonously increasing throughout the whole Sphere. Our results cannot exclude that the ratio can reach  $R(\pi; t) > 1$  in the continuum limit. For large  $n \gg 1$  adding one more convolution will not produce any appreciable changes to the pair density matrix of Eq. (2.12), or in other words  $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; T)$  will be to a good approximation a fixed point such that

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; T) = \int \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}''_1, \mathbf{r}''_2; T) \rho(\mathbf{r}'_1, \mathbf{r}'_2 | \mathbf{r}''_1, \mathbf{r}''_2; \tau) \sqrt{g(\mathbf{r}''_1)g(\mathbf{r}''_2)} d\mathbf{r}''_1 d\mathbf{r}''_2, \quad (2.23)$$

with  $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau)$  the primitive approximation (2.14) to the high temperature density matrix. As we already discussed the fixed point will be a stable sink if the pair potential is bounded from below. Therefore we expect that the pair-inter-action (2.10) will approach the fixed point at  $t = n\tau$  and change less and less as  $n$  increases.

The central panel of Fig. 1 shows the dependence of the ratio  $R$  on the choice of the Coulomb potential just for  $n = 2$ .

The bottom panel of the figure shows the evolution of the peak for  $n = 2$  at different values of the Sphere radius. It is important to note that the peak is an effect of the curvature of the surface and we explicitly verified that for  $n = 2$  it disappears on a periodic flat square where the ratio is a monotonous increasing function of the distance reaching  $\approx 1$  at the edge of the square box. This also shows the difference between a Sphere of infinite radius and a periodic flat square.

In Table I we show  $P(0; t)$  and  $R(0; t)$  for  $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2$  and increasing  $n$ . The value at contact,  $d_g = 0$ , is obtained through a linear extrapolation of the first two points at  $m = 1, 2$  of the top panel of Fig. 1. We see how  $R(0; t)$  decreases approximately as  $1/n$  at increasing  $n$ . Therefore we expect that at constant  $t = n\tau$  as  $n \rightarrow \infty$ , in the  $\tau \rightarrow 0$  continuum limit, the ratio  $R(0; t) \rightarrow 0$ . Then we can say that  $P(d_g \rightarrow 0; t)$  must be an infinite of order less than the infinite of  $\phi(d_g \rightarrow 0)$ . Moreover the PCDM at particles contact must obey to a cusp condition [6] as discussed in the next section.

TABLE I. Table for  $P_{\text{on}}^g(0; t)$  and  $R_{\text{on}}^g(0; t) = \lim_{\delta \rightarrow 0} P_{\text{on}}^g(\delta; t)/t\phi_{\text{on}}^g(\delta)$  for  $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2$  and increasing  $n$ . The value at contact,  $d_g = 0$ , is obtained through a linear extrapolation of the first two points at  $m = 1, 2$  of the top panel of Fig. 1.

| $n$ | $\tau$ | $P_{\text{on}}^g(0; 0.2)$ | $R_{\text{on}}^g(0; 0.2)$ |
|-----|--------|---------------------------|---------------------------|
| 2   | 0.1000 | 5.50                      | 0.517                     |
| 3   | 0.0666 | 4.31                      | 0.364                     |
| 4   | 0.0500 | 3.50                      | 0.250                     |
| 5   | 0.0400 | 3.10                      | 0.207                     |
| 6   | 0.0333 | 3.40                      | 0.162                     |

In Appendix D we present the FORTRAN code used to generate Fig. 1.

### III. CUSP CONDITION

On the sphere the cusp condition [1, 5, 6]. can be determined studying the asymptotic behavior as  $\mathbf{r}_1 \rightarrow \mathbf{r}_2$  of a vanishing *residual energy* (defined in equation (4.6) of Ref. [1]) for the pair-product approximation to the inter-action.

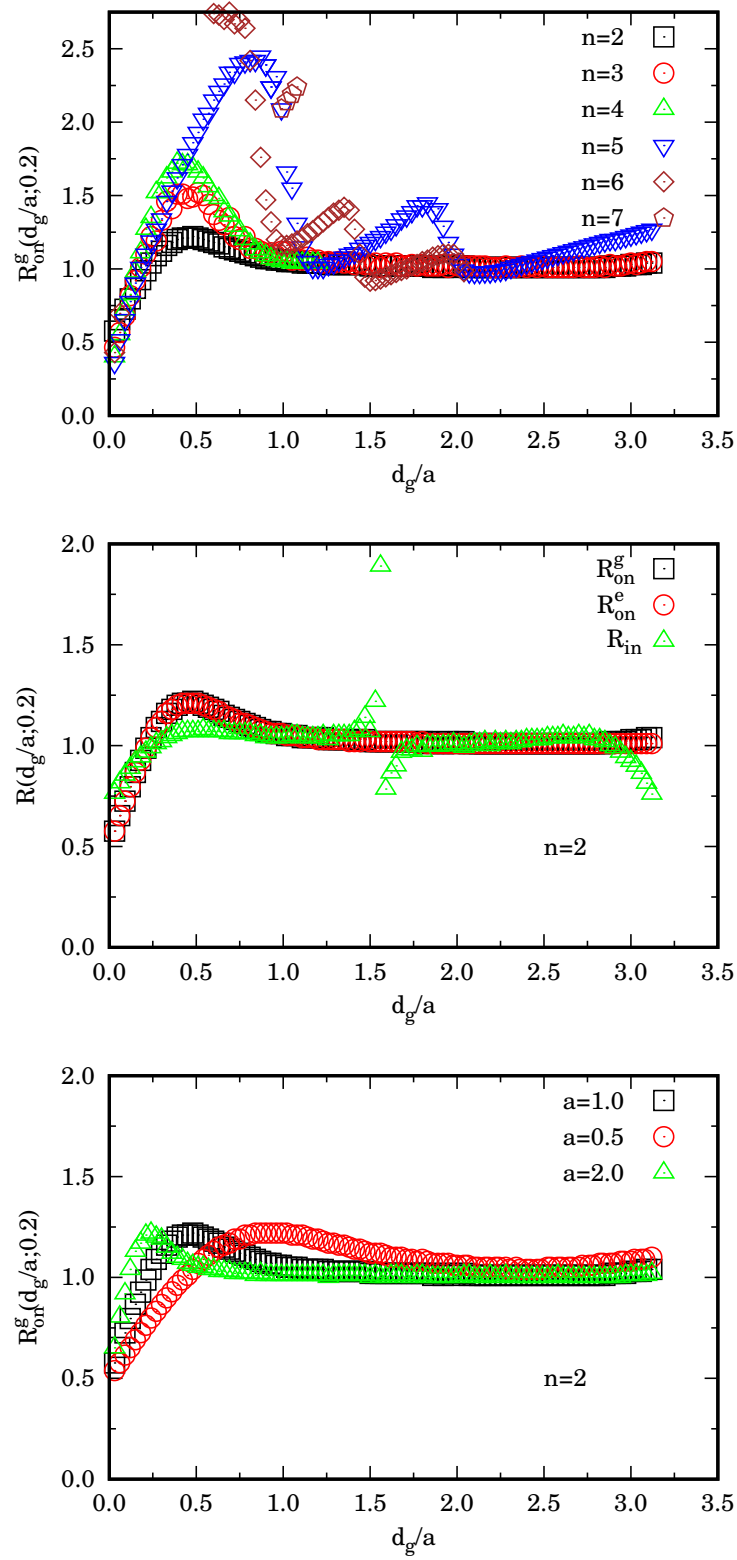


FIG. 1. We show the plot of the ratio  $R(d_g/a; t) = P(m\delta; t)/t\phi(m\delta)$  as a function of  $d_g = m\delta$  for  $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2, \delta = 0.03$ . On the top panel  $R_{on}^g$  for  $n = 2, 3, 4, 5, 6, 7$ . For the cases  $n = 6, 7$  we show only a few points. On the central panel the three Coulomb cases  $R_{on}^g, R_{on}^e, R_{in}$  for  $n = 2$ . On the bottom panel the case  $R_{on}^g$  for  $n = 2$  and different Sphere radii. The angular convolutions in Eq. (2.12) were calculated with a Monte Carlo quadrature with no less than  $10^6$  MC points and keeping the relative statistical error below 10% (this required approximately  $> 10^{10}$  MC points in the  $n > 4$  cases).

On a Sphere as  $\mathbf{r}_1 \rightarrow \mathbf{r}_2$  we should compare

$$\left( \frac{\lambda_1}{\sqrt{g(\mathbf{r}_1)}} \nabla_{1\alpha} \sqrt{g(\mathbf{r}_1)} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g(\mathbf{r}_2)}} \nabla_{2\alpha} \sqrt{g(\mathbf{r}_2)} \nabla_2^\alpha \right) P(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t) = -\phi(\mathbf{r}_1, \mathbf{r}_2). \quad (3.1)$$

As  $\mathbf{r}_1 \rightarrow \mathbf{r}_2$ ,  $d_g(\mathbf{r}_1, \mathbf{r}_2) \rightarrow 0$  and we are looking at a local property on the sphere so that we can use the result on the tangent plane at the point  $\mathbf{r}_1 = \mathbf{r}_2$  namely, for any  $t$ ,

$$\lim_{d_g \rightarrow 0} \frac{\partial P_{\text{on}}(d_g/a; t)}{\partial d_g} = \frac{Z_1 Z_2}{\lambda_1 + \lambda_2}, \quad (3.2)$$

$$\lim_{d_g \rightarrow 0} \frac{\partial P_{\text{in}}(d_g/a; t)}{\partial d_g} = 0, \quad (3.3)$$

where we used the following expression for the Laplacian in an Euclidean space of dimension  $d$  in an isotropic state

$$\nabla_r^2 = \frac{\partial^2}{\partial r^2} + \frac{d-1}{r} \frac{\partial}{\partial r}. \quad (3.4)$$

#### IV. CONCLUSIONS

We tabulated the (end-point approximation) diagonal inter-action for the repulsive  $Z_1 Z_2 > 0$  pair Coulomb density matrix on the Sphere at a fixed temperature  $T_0 = T_n/n$  obtained from the classical high temperature primitive approximation through a discretized path integral calculation with imaginary timestep  $\tau = 1/T_n \rightarrow 0$  in the continuum  $n \rightarrow \infty$  limit. We do this from a path integral calculated through a Monte Carlo quadrature. This only allows the determination of the repulsive ( $Z_1 Z_2 > 0$ ) pair-inter-action because in the attractive ( $Z_1 Z_2 < 0$ ) case the primitive pair-inter-action, i.e. the bare Coulomb pair-potential, is unbounded from below and Trotter formula breaks down. We took care explicitly of the  $n = 2$  square, the  $n = 3$  cube, up to  $n = 6$  case. Higher  $n$  cases become progressively more and more computationally demanding. We found that the ratio between the diagonal inter-action and its primitive approximation tends to  $\approx 1/n$  at particles contact. This suggests that it will go to zero in the continuum limit  $n \rightarrow \infty$  for a fixed temperature. If, for the case of two particles on a periodic square on a plane, the ratio have a monotonously increasing behavior from 0 to 1 for all  $n$ , on the Sphere the ratio has a monotonous behavior only in the continuum limit and it can reach a value bigger than 1 at the maximum geodesic distance between the two particles.

We also derive the cusp condition for the pair Coulomb density matrix on the Sphere. This has the same form as in flat space since as the two particles comes together at a point  $\mathbf{r}$  their diagonal pair Coulomb density matrix reduces to the one on the tangent plane at  $\mathbf{r}$ .

For any finite  $n$  one can treat the attractive  $Z_1 Z_2 < 0$  Coulomb component provided some caution is taken numerically (see Appendix D). This amounts to introduce a numerical cutoff, imposing  $d_g(\mathbf{r}_1, \mathbf{r}_2) > \sigma$ , or, in physical terms, a hard core between the two unlike particles. For the repulsive component we do not have this problem. We showed, numerically on a Sphere, the progressive smearing property of the divergence of the pair-inter-action of the primitive approximation obtained taking the  $n$ th powers of the primitive approximation for increasing  $n$ . Our results confirm how quantum statistical physics is fundamentally different from classical statistical physics: if in the latter one needs to artificially introduce a regularization of the divergent Coulomb potential preventing particles contact, for example with a hard core, in the former this is not necessary since the divergences in the Coulomb pair-inter-action are washed out.

If one is looking for the exact attractive  $Z_1 Z_2 < 0$  component of the pair Coulomb density matrix it is not possible to derive it through a path integral from the primitive approximation although one can gain numerically some insight by progressively reducing the extent  $\sigma$  of the hard core and increasing the number  $n$  of timeslices.

Even if we are still far from an exact analytic solution of the Hydrogen atom problem on the surface of a Sphere with the path integral method, akin to what was done in Euclidean space in Ref. [7], nonetheless we found some numerical results that can give some insight in the search for a solution of the problem. We plan, in the future, to find the diagonal pair-inter-action near the continuum limit  $n \gg 1$ , up to an additive constant, using the path integral Monte Carlo method [1] to calculate the histogram

$$\rho(d/a; n\tau) \propto \langle \delta[d_g(\mathbf{r}_1, \mathbf{r}_2) - d] \rangle_T, \quad (4.1)$$

where  $T = n\tau$ ,  $\delta[\dots]$  is a Dirac delta function, and  $\langle \dots \rangle$  denotes an average with respect to the diagonal pair density matrix (2.12) at a temperature  $T$ . Even though this calculation is able to determine the behavior near the continuum limit it cannot find the additive constant.

### Appendix A: Coulomb potential in the sphere surface

Place one particle on the north pole  $\theta = 0$  so that the geodesic and Euclidean distances between the two particles are respectively  $d_g = a\theta$  and  $d_e = 2a \sin(\theta/2)$ . The Poisson equation in the sphere surface is

$$\Delta G = -2\pi\delta^{(2)}, \quad (\text{A1})$$

where  $\Delta = g^{-1/2}\nabla_\alpha g^{1/2}g^{\alpha\beta}\nabla_\beta$  is the Laplace-Beltrami operator,  $\delta^{(2)} = g^{-1/2}\delta(\theta)\delta(\varphi)$  is the Dirac delta, and  $G$  is the Coulomb potential.

Due to rotational symmetry around the  $z$  axis, Poisson equation reduces to

$$\partial_\theta \sin\theta \partial_\theta G(\theta) = -2\pi\delta(\theta). \quad (\text{A2})$$

For  $\theta \neq 0$  a solution is

$$G(\theta) = -C \ln[\tan(\theta/2)], \quad (\text{A3})$$

To determine the constant  $C$  we integrate Poisson equation (A1) over the sphere surface

$$-2\pi = \int \Delta G g^{1/2} dS = 2\pi \left( g^{1/2}\nabla^\theta G \Big|_{\theta=\pi} + g^{1/2}\nabla^\theta G \Big|_{\theta=0} \right) = -4\pi C, \quad (\text{A4})$$

where  $dS = d\theta d\varphi$  and  $g^{1/2}\nabla^\theta G = -C$ .

Then

$$G(\theta) = -\frac{1}{2} \ln[\tan(\theta/2)]. \quad (\text{A5})$$

### Appendix B: Eigenfunctions of the Hamiltonian (2.2) for $\lambda_1 \rightarrow 0$

If  $m_1 \rightarrow \infty$  we can easily find the eigenfunctions of the Hamiltonian (2.2) by fixing particle 1 at the north pole  $\theta_1 = 0$ . Then Shrödinger equation  $\mathcal{H}\psi_n = E_n\psi_n$  for the eigenfunctions  $\psi_n(\theta_2, \varphi_2)$  becomes

$$\left\{ -\frac{\lambda_2}{a^2} \left[ \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left( \sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right] + \phi(\theta) \right\} \psi_n = E_n\psi_n, \quad (\text{B1})$$

where  $\theta = \theta_2$ ,  $\varphi = \varphi_2$ , and the Coulomb potential

$$\phi_{\text{on}}^g(\theta) = Z_1 Z_2 e^2 / a\theta, \quad (\text{B2})$$

$$\phi_{\text{on}}^e(\theta) = Z_1 Z_2 e^2 / a\sqrt{2(1 - \cos\theta)}, \quad (\text{B3})$$

$$\phi_{\text{in}}(\theta) = -Z_1 Z_2 e^2 \ln \left[ \sqrt{2(1 - \cos\theta)} a / \ell \right]. \quad (\text{B4})$$

Since the Coulomb potential depends only on  $\theta$  the azimuthal angle separates and we can look for eigenfunction of the following kind:  $\psi(\theta, \varphi) = \Theta(\theta)e^{\pm im\varphi}$ , where  $m$  is an integer, the magnetic quantum number. Substituting this into the Schrödinger equation (B1) gives an ordinary differential equation in only  $\theta$ . It is convenient the following change of variable  $x = \cos\theta$ . Then we have to solve

$$\left\{ -\frac{\lambda_2}{a^2} \left[ \frac{d}{dx} (1-x^2) \frac{d}{dx} - \frac{m^2}{1-x^2} \right] + \phi(x) \right\} \Theta(x) = E\Theta(x), \quad (\text{B5})$$

where we are looking for a solution  $\Theta$  which remains finite for  $x^2 \leq 1$ . For  $\phi = 0$  the solution is the usual one in terms of Legendre polynomials. For the Coulomb potentials of Eqs. (B2)-(B4) one should look for a numerical solution. The case of a finite sphere radius  $a$  shows the effect of curvature on the case of the plane  $a \rightarrow \infty$ .



### Appendix C: Diagonal free particle density matrix

Calling  $\mu_i = m_i/m_e$  and  $R = 2/a^2$  the Sphere scalar curvature, we find from equation (3.23) of Ref. [13]

$$\begin{aligned} \ln \left[ 2\pi t \sqrt{g(\mathbf{0})} \rho_{i0}(\mathbf{0}|\mathbf{0}; t) / \mu_i \right] = & \frac{tR}{12} + \\ & \frac{(tR)^2}{6!} \frac{1}{2} + \\ & \frac{(tR)^3}{9!} 16 + \\ & \frac{(tR)^4}{10!} \frac{59}{4} + \\ & \frac{(tR)^5}{11!} \frac{58}{3} + \\ & \frac{(tR)^6}{13!} \frac{550789}{1260} + \\ & \frac{(tR)^7}{14!} \frac{46838}{45} + \\ & \frac{(tR)^8}{17!} \frac{596004707}{720} + \mathcal{O}(t^9), \end{aligned} \tag{C1}$$

where due to the Sphere isotropy  $\sqrt{g(\mathbf{0})} \rho_{i0}(\mathbf{0}|\mathbf{0}; t) = \sqrt{g(\mathbf{r})} \rho_{i0}(\mathbf{r}|\mathbf{r}; t)$  for any  $\mathbf{r}$ .

### Appendix D: The code

We give here the FORTRAN code listing to generate the table of Eq. (2.22). To tabulate the unlike attractive  $Z_1 = -Z_2$  case it is necessary to choose `epsd` not too small, say 0.1 in order to prevent the overflow of the exponential of the attractive Coulomb potential when  $d_g(\mathbf{r}_1, \mathbf{r}_2) < \text{epsd}$ .



```

end

function pot(potk,a,z1,z2,dg,de)
implicit none
c the Coulomb potential
real*8 pot,z1,z2,a
real*8 de,dg
character potk*3

if(potk.eq.'in') then
  pot=-z1*z2*log(tan(dg/2/a))/2
elseif(potk.eq.'one') then
  pot=z1*z2/de
elseif(potk.eq.'ong') then
  pot=z1*z2/dg
else
  write(*,*) 'no pair-potential available !'
  stop
endif

return
end

function rho(potk,tau,a,mu1,mu2,z1,z2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p,dg,de,dgp,dep)
implicit none
c primitive approximation for rho
real*8 rho,tau,a,mu1,mu2,z1,z2,dg,de,dgp,dep
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 pot,vr,ddx1,ddx2
character potk*3

ddx1=(a*th1*cos(ph1)-a*thip*cos(phip))*2.+
& (a*th1*sin(ph1)-a*thip*sin(phip))*2.
ddx2=(a*th2*cos(ph2)-a*th2p*cos(ph2p))*2.+
& (a*th2*sin(ph2)-a*th2p*sin(ph2p))*2.

rho = exp(-mu1*ddx1/2/tau-tau*
& (vr(a,mu1,th1)+vr(a,mu1,thip))/2)
rho=rho*exp(-mu2*ddx2/2/tau-tau*
& (vr(a,mu2,th2)+vr(a,mu2,th2p))/2)
rho=rho*exp(-tau*(pot(potk,a,z1,z2,dg,de)+
& pot(potk,a,z1,z2,dgp,dep))/2)

return
end

function rhoid(tau,a,mu1,mu2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p)
implicit none
c primitive approximation for ideal gas rho
real*8 rhoid,tau,a,mu1,mu2
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 vr,ddx1,ddx2

ddx1=(a*th1*cos(ph1)-a*thip*cos(phip))*2.+
& (a*th1*sin(ph1)-a*thip*sin(phip))*2.
ddx2=(a*th2*cos(ph2)-a*th2p*cos(ph2p))*2.+
& (a*th2*sin(ph2)-a*th2p*sin(ph2p))*2.

rhoid = exp(-mu1*ddx1/2/tau-tau*
& (vr(a,mu1,th1)+vr(a,mu1,thip))/2)
rhoid=rhoid*exp(-mu2*ddx2/2/tau-tau*
& (vr(a,mu2,th2)+vr(a,mu2,th2p))/2)

return
end

function rho0(a,mu,t)
implicit none
c EQ. (3.23) Bastianelli and Corradini EPJC 77, 731 (2017)
real*8 rho0,a,mu,t,th
real*8 pi,fact,rit
real*8 c1,c2,c3,c4,c5,c6,c7,c8
parameter(c1=1./2.,c2=16./4.,c3=59./4.,c4=58./3.,c5=550789./1260,
& c6=46838./45.,c7=596004707./720)
pi=acos(-1.d0)

rit=t*2./a**2.
rho0=rit/12*c1*rit**2./fact(6)+c2*rit**3./fact(9)+
& c3*rit**4./fact(10)+c4*rit**5./fact(11)+c5*rit**6./fact(13)+
& c6*rit**7./fact(14)+c7*rit**8./fact(17)
rho0=mu*exp(rho0)/(2.*pi*t)

return
end

```

## AUTHOR DECLARATIONS

### Conflicts of interest

None declared.

### Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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